Supersymmetric Gelfand-Dickey Algebra

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Abstract

We study the classical version of supersymmetric W-algebras. Using the second Gelfand-Dickey Hamiltonian structure we work out in detail W_2 and W_3-algebras.

1 Introduction

.30in K. Huitu math expression Supported by a grant from the Finnish Cultural Foundation and D. Nemeschansky math expression Supported by the Department of Energy grant DE-FG03-84ER-40168 Department of Physics University of Southern California Los Angeles, CA 90089 .3in We study the classical version of supersymmetric math expression -algebras. Using the second Gelfand-Dickey Hamiltonian structure we work out in detail math expression and math expression -algebras.

2 Method

Above we have used math expression for multiplication to distinguish math expression from math expression . The coefficients math expression in eq. \ are the superbinomial coefficients, j k \ = \ & math expression for math expression and for math expression mod math expression & math expression for math expression , math expression mod math expression \ \ ,

$$L(\theta) = \sum_{i=1}^{n} \ell(x_i, \theta) \tag{1}$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

& f(z,) exp (- 2 n ^z u_ n-1 (z_1,) dz_1) f(z,) & f(z,) exp (- 2 n-1 ^z ^ z^ D_ z_1, u_ n-1 (z_1,) dz_1 dz^) f(z,) \ . we can eliminate the coefficient math expression from the expansion of the differential operator . Therefore the differential operator math expression has the form L \ = \ D^n + u_ n-2 (z,)D^ n-2 + + u_ 1 (z,) D + u_0 \ \ . Next let us consider the behavior of the differential operator under a superconformal transformation. A super analytic map math expression transforms the superderivative according to, e.g. \